Lab 04

COMP1323 Networks and Security

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Activity 1

i) Create four subnets from your assigned network to support the four new computer laboratories.

Class A:
10.19.0.0/8 -> N.H.H.H
Class B:
172.19.0.0/16 -> N.N.H.H
Class C:
192.168.19.0/24 -> N.N.N.H
For 30 workstations in IoT Lab, we need exactly 5 host bits because 2^5 - 2 = 30 (which is equal to 30). This perfectly fits the requirement. This leaves 32 - 5 = 27 network bits.
For the same 30 workstations in Security Lab, we also need 5 host bits because 2^5 - 2 = 30. This also leaves 32 -5 =27 network bits.
For 45 workstations in Networking Lab, 5 host bits (2^5 - 2 = 30) is not enough. We still need 6 host bits (2^6 - 2 = 62 (which is bigger than 45)). This leaves 26 network bits.
For 60 workstations in the General Purpose Lab, we need a subnet size that has at least 60 usable IP addresses. The next power of 2 is 64 (2^64). A block size of 64 requires 6 host bits, leaving 26 network bits. Hence, we can divide the **/24** network into four equal **/26** networks.

ii) Determine the subnet mask for each subnet. (3 Marks)

Class A:
IoT lab: **10.19.0.128/27 (255.255.255.0)**
Security Lab: **10.19.0.160/27**
Networking Lab: **10.19.0.64/26**
General Purpose Lab: **10.19.0.0/26**
Class B:
IoT lab: **172.19.0.128/27**
Security Lab: **172.19.0.160/27**
Networking Lab: **172.19.0.64/26**
General Purpose Lab: **172.19.0.0/26**
Class C:
IoT lab: **192.168.19.128/27** (which is **255.255.255.224** in decimal)
Security Lab: **192.168.19.160/27** (**255.255.255.224** in decimal)
Networking Lab: **192.168.19.64/26** (**255.255.255.192** in decimal).
General Purpose Lab: **192.168.19.0/26** (**255.255.255.192** in decimal).

iii) For each subnet, calculate: (8 Marks)

- **Network Address**
- **First usable host address**
- **Last usable host address**

• **Broadcast address**

Lab Name	Subnet Mask	Network Address	First Usable Host	Last Usable Host	Broadcast Address
IoT Lab	/27	192.168.19.128	192.168.19.129	192.168.19.158	192.168.19.159
Security Lab	/27	192.168.19.160	192.168.19.161	192.168.19.190	192.168.19.191
Networking Lab	/26	192.168.19.64	192.168.19.65	192.168.19.126	192.168.19.127
General Purpose Lab	/26	192.168.19.0	192.168.19.1	192.168.19.62	192.168.19.63

iv) Briefly explain two benefits of subnetting in a university network. (2 Marks)
First of all, it will reduce unnecessary network traffic and congestion by dividing the large broadcast domain into smaller ones, thus a small university just like the University of Southampton Malaysia campus can have better network performance. Secondly, it will allow us to isolate our sensitive networks, like the Security Lab, from the general student networks, like the General Purpose Lab, by using access control lists on the router.

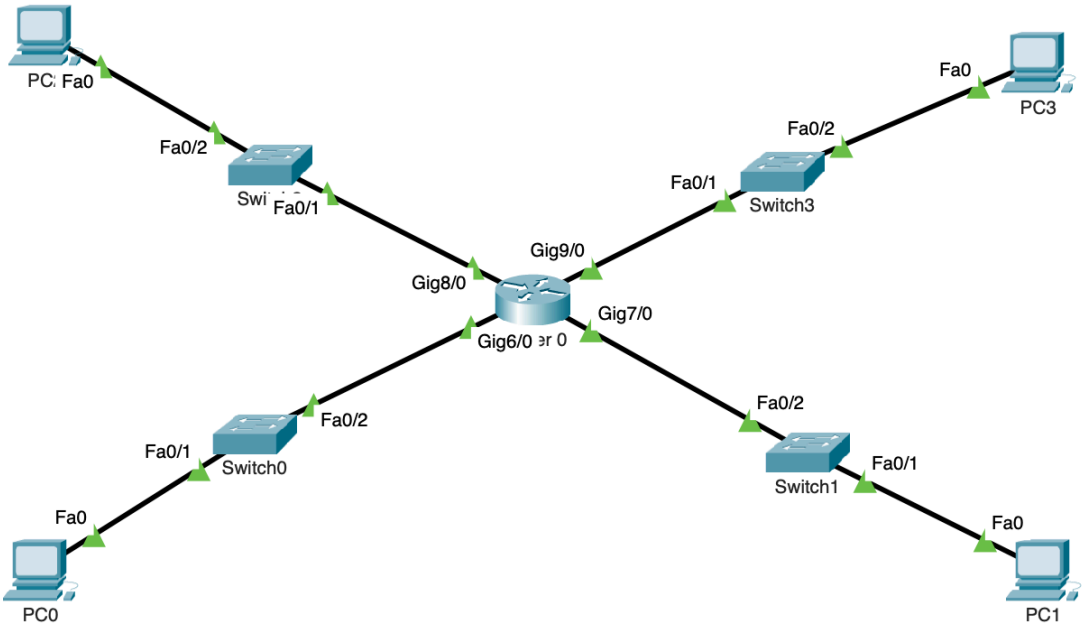
Activity 2:

i. Network Topology Design

1. Place 1 Router (e.g., 2911) in the center.
2. Place 4 Switches (e.g., 2960) around the router, one for each lab.
3. Place 4 PCs (PC1, PC2, PC3, PC4) and connect one PC to each switch.

4. Use Copper Straight-Through Cables to connect the PCs to the Switches, and the Switches to the GigabitEthernet interfaces on the Router.

Screenshot:



[1]

ii. IP Address Configuration (4 Marks)

Configure IP addresses according to your subnetting design:

• Assign IP addresses to PCs

• Configure router interfaces

• Set the default gateway for each PC

Students must use the network addresses assigned based on their Student ID.

Take screenshots of IP Address Configuration.

iii. Connectivity Testing (4 Marks)

Verify connectivity using the Ping command.

Test the following:

• PC1 → PC2

• PC1 → PC3

• PC1 → PC4

Take screenshots of successful ping results.

Router Configuration (Default Gateways)

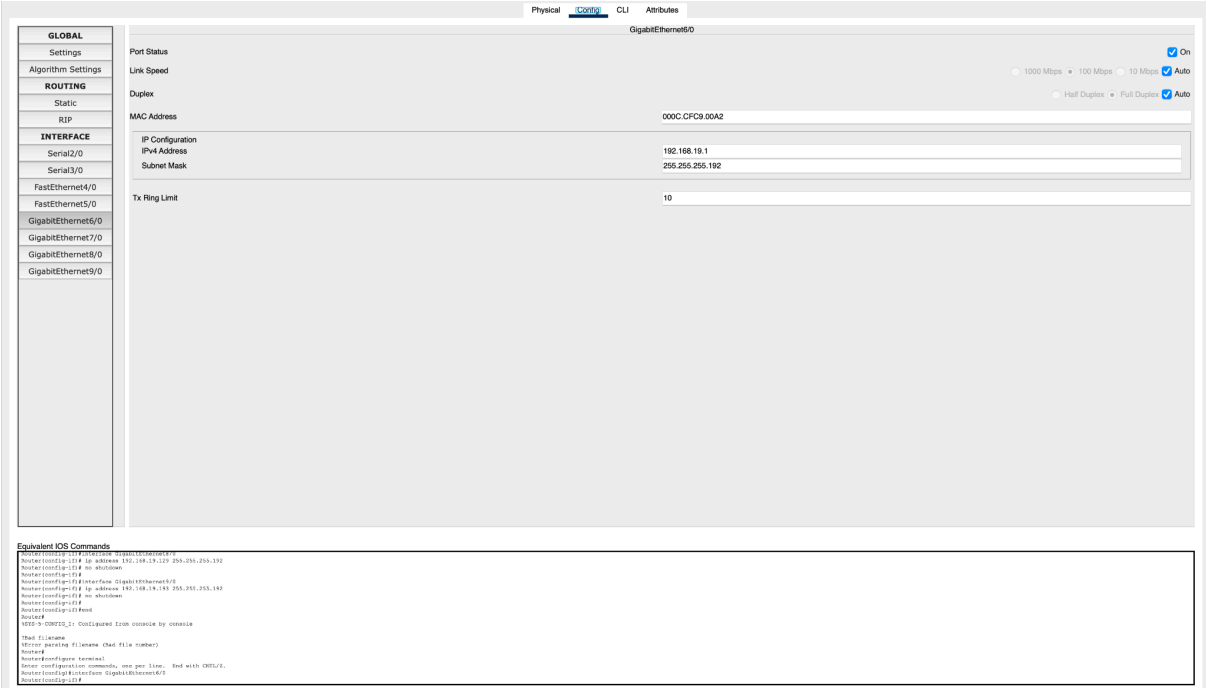
Interface	Lab	IP Address	Subnet Mask

[1] Screenshot of Network Topology Design

GigabitEthernet6/0	General Lab	192.168.19.1	255.255.255.192
GigabitEthernet7/0	Networking	192.168.19.65	255.255.255.192
GigabitEthernet8/0	IoT	192.168.19.129	255.255.255.192
GigabitEthernet9/0	Security	192.168.19.193	255.255.255.192

Screenshot:

GigabitEthernet6/0:



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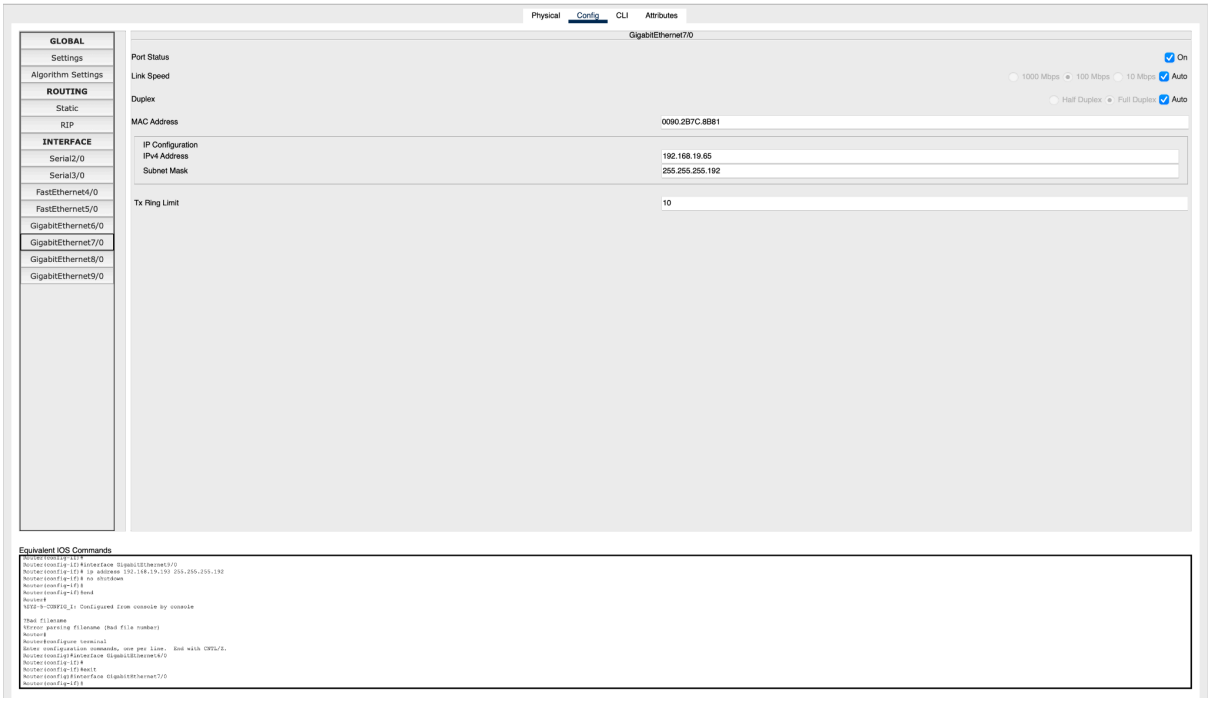
GigabitEthernet7/0:

[2] Screenshot of GigabitEthernet6/0 IP config



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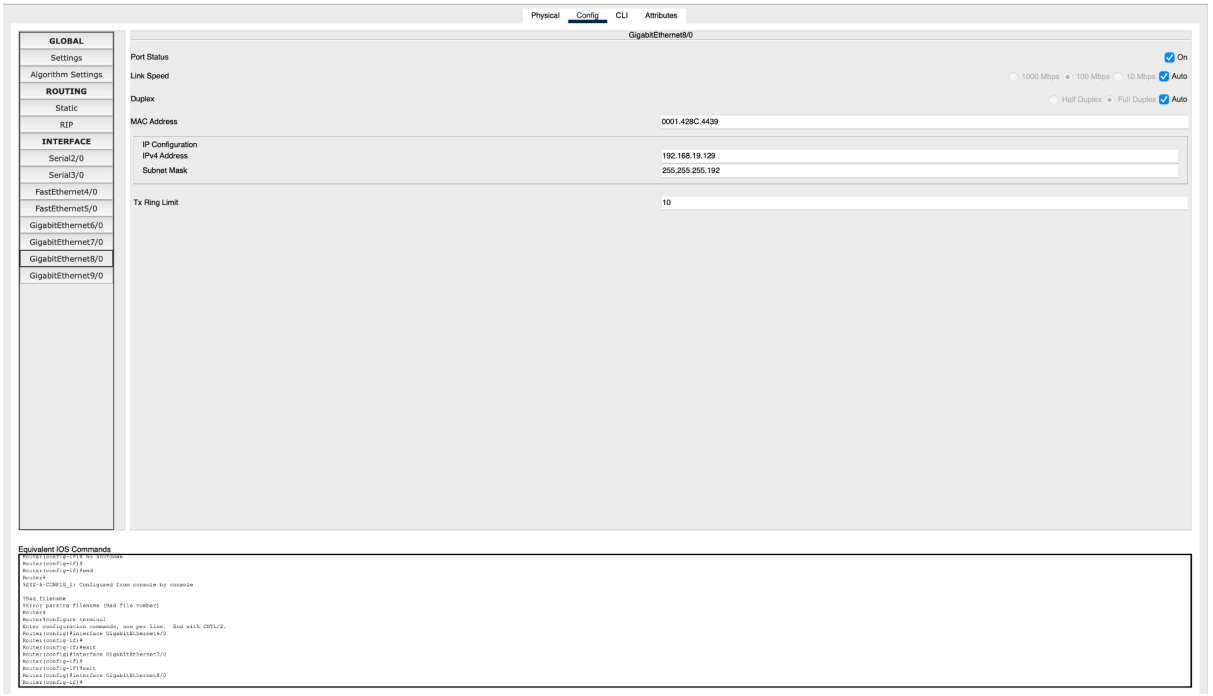
GigabitEthernet8/0:



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GigabitEthernet9/0:

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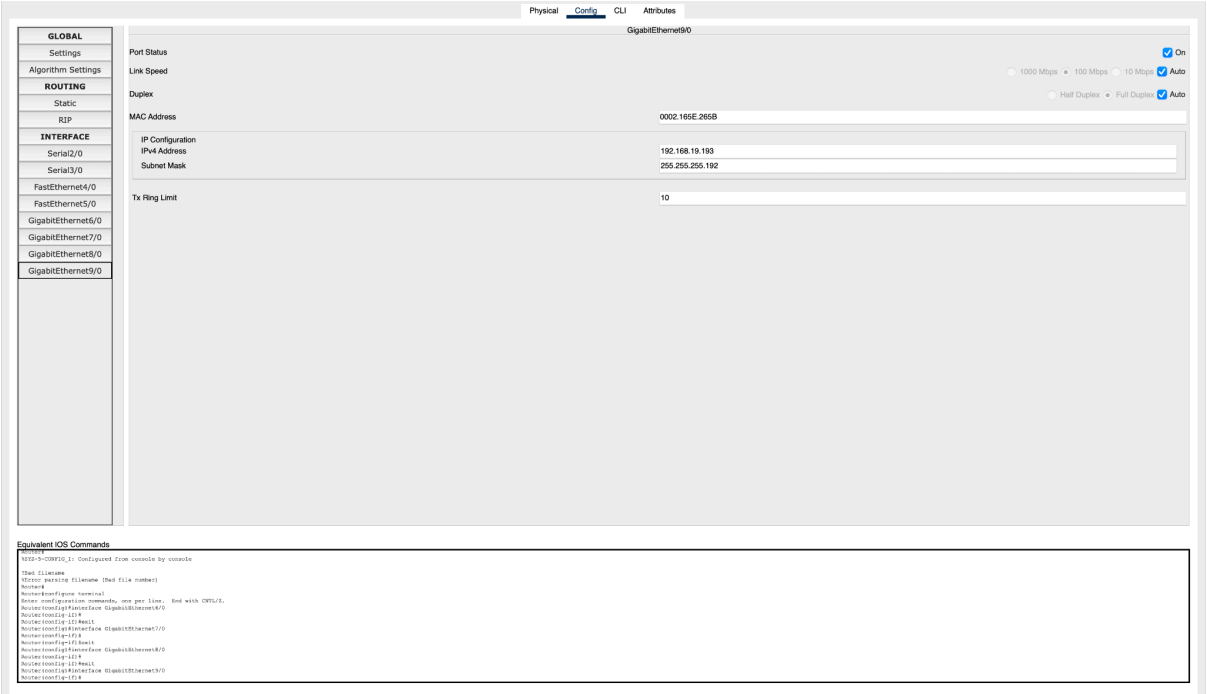
[3] Screenshot of GigabitEthernet7/0 IP config
[4] Screenshot of GigabitEthernet8/0 IP config



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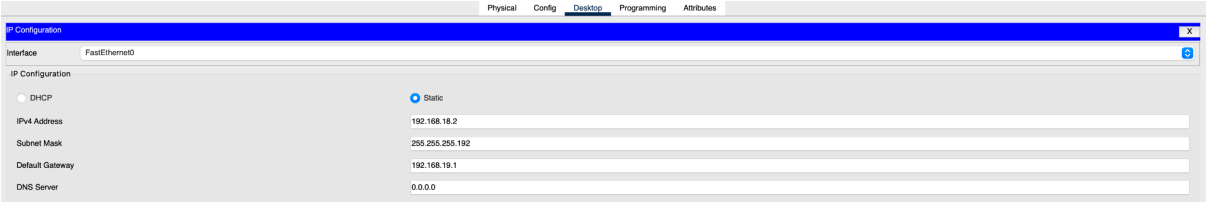
End Device (PC) Configuration

Device	Department	IP Address	Subnet Mask	Default Gateway
PC0	General	192.168.19.2	255.255.255.192	192.168.19.1
PC1	Networking	192.168.19.66	255.255.255.192	192.168.19.65
PC2	IoT	192.168.19.130	255.255.255.192	192.168.19.129
PC3	Security	192.168.19.194	255.255.255.192	192.168.19.193

Screenshot:

General Lab:

[5] Screenshot of GigabitEthernet9/0 IP config



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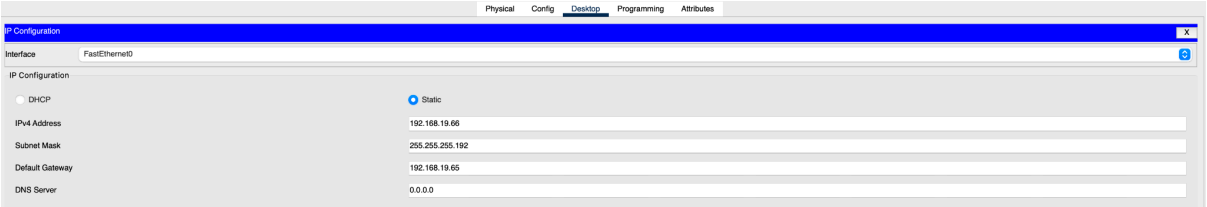
Networking Lab:



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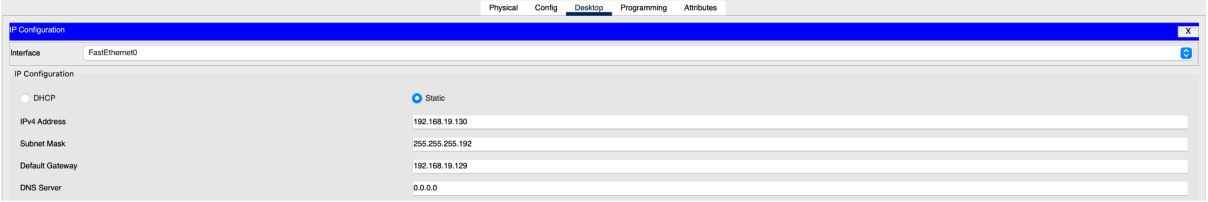
IoT Lab:



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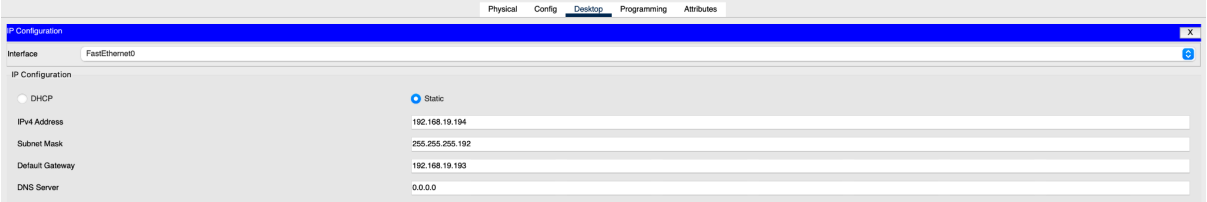
Security Lab:



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iii. Connectivity Testing

The packet loss occurs due to ARP resolution. The first ICMP packet is dropped while the device resolves the MAC address, which is quite normal in network communication.

Ping test on Router:

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[6] Screenshot of General Lab PC0 IP config
[7] Screenshot of Networking Lab PC1 IP config
[8] Screenshot of IoT Lab PC2 IP config
[9] Screenshot of Security Lab PC3 IP config

```
Router#ping 192.168.19.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.19.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.19.66

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.19.66, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.19.130

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.19.130, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.19.194

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.19.194, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

[10]Ping test on PC 0:

[10] Screenshot of PING test on Router

```
C:\>ping 192.168.19.1

Pinging 192.168.19.1 with 32 bytes of data:

Reply from 192.168.19.1: bytes=32 time<1ms TTL=255
Reply from 192.168.19.1: bytes=32 time<1ms TTL=255
Reply from 192.168.19.1: bytes=32 time<1ms TTL=255
Reply from 192.168.19.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.19.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.19.66

Pinging 192.168.19.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.19.66: bytes=32 time<1ms TTL=127
Reply from 192.168.19.66: bytes=32 time<1ms TTL=127
Reply from 192.168.19.66: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.19.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.19.130

Pinging 192.168.19.130 with 32 bytes of data:

Request timed out.
Reply from 192.168.19.130: bytes=32 time<1ms TTL=127
Reply from 192.168.19.130: bytes=32 time<1ms TTL=127
Reply from 192.168.19.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.19.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.19.194

Pinging 192.168.19.194 with 32 bytes of data:

Request timed out.
Reply from 192.168.19.194: bytes=32 time<1ms TTL=127
Reply from 192.168.19.194: bytes=32 time<1ms TTL=127
Reply from 192.168.19.194: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.19.194:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```
[11]

[11] Screenshot of PING test on PC0

Activity 3: Router Packet Scheduling

i. First Come First Serve (FCFS)

Gantt Chart:

| P1 (0-5) | P2 (5-8) | P3 (8-12) | P4 (12-14) |

Waiting Times (Start Time - Arrival Time):

- P1: 0 - 0 = 0 ms
- P2: 5 - 1 = 4 ms
- P3: 8 - 2 = 6 ms
- P4: 12 - 3 = 9 ms
- Average Waiting Time: (0 + 4 + 6 + 9) / 4 = 4.75 ms

ii. Priority Scheduling (Non-Preemptive)

Gantt Chart:

| P1 (0-5) | P2 (5-8) | P3 (8-12) | P4 (12-14) |

- **P1:** 0 - 0 = **0 ms**
- **P2:** 5 - 1 = **4 ms** (Waited from 1ms to 5ms)
- **P3:** 8 - 2 = **6 ms** (Waited from 2ms to 8ms)
- **P4:** 12 - 3 = **9 ms** (Waited from 3ms to 12ms)
- Average Waiting Time: (0 + 4 + 6 + 9) / 4 = = 4.75 ms

iii. Round Robin (Time Quantum = 2 ms)

Gantt Chart:

| P1(0-2) | P2(2-4) | P3(4-6) | P4(6-8) | P1(8-10) | P2(10-11) | P3(11-13) | P1(13-14) |

Waiting Times (Turnaround Time - Processing Time):

- P1: (Finished at 14) Turnaround: 14 - 0 = 14. Wait: 14 - 5 = 9 ms
- P2: (Finished at 11) Turnaround: 11 - 1 = 10. Wait: 10 - 3 = 7 ms
- P3: (Finished at 13) Turnaround: 13 - 2 = 11. Wait: 11 - 4 = 7 ms
- P4: (Finished at 10) Turnaround: 8 - 3 = 5. Wait: 5 - 2 = 3 ms
- Average Waiting Time: (9 + 7 + 7 + 3) / 4 = 6.5 ms

Activity 4: Shortest Path Routing using Dijkstra's Algorithm

i. Step-by-step table starting from Router A

Iteration	Set of Visited Nodes (N')	D(B)	D(C)	D(D)	D(E)	D(F)
0	{A}	**3 (A)**	5 (A)	9 (A)	∞	∞
1	{A, B}	3	**5 (A)**	7 (B)	10 (B)	∞
2	{A, B, C}	3	5	**7 (B)**	10 (B)	13 (C)
3	{A, B, C, D}	3	5	7	**9 (D)**	9 (D)
4	{A, B, C, D, E}	3	5	7	9	**9 (D)**
5	{A, B, C, D, E, F}	3	5	7	9	9

ii. Shortest path from Router A to Router F Tracing back from F:

F connects to D (cost 2). D connects to B (cost 4). B connects to A (cost 3).

Path: A –---> B –->D –->F (Total Cost = 9)

(Same for A —>C —>D —>F ,and this total cost is also 9).

iii. Final Routing Table for Router A

Destination Network	Next Hop	Total Cost
A	-	0
B	B	3
C	C	5
D	B (or C)	7
E	B (or C)	9
F	B (or C)	9